Master Thesis

Investigating Ensemble Strategies for Enhancing the Generalization of Foundation Models in Outlier Detection

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Introduction

Motivation

- **Outlier detection (OD)** in tabular data has numerous applications in critical domains such as security, environmental monitoring, finance, etc.
- Deep OD models have not yet achieved a significant breakthrough compared to classical methods.
- The recent success of **large foundation models** offers new opportunities for **zero-shot OD**.
- **FoMo-0D** [Shen et al.([n. d.])], the first foundation model for zero-shot OD on unseen tabular datasets, demonstrates this potential.

Motivation

$\Rightarrow$ Our work represents the first study investigating **ensemble strategies based on the FoMo-0D architecture** for tabular outlier detection.

Related Work

FoMo-0D is a pre-trained **Fo**undation **Mo**del for zero-shot/**0**-shot O**D** on tabular data [Shen et al.([n. d.])].

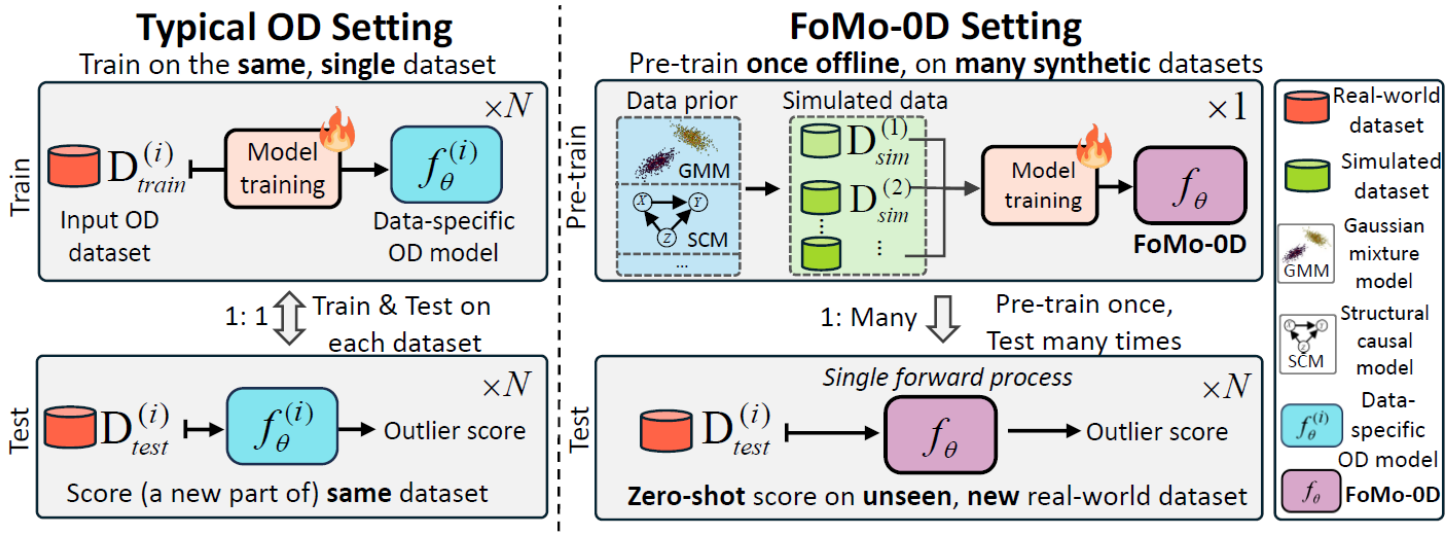


Figure 1: Comparison of typical OD vs. the FoMo-0D settings [Shen et al.([n. d.])]

Methodology

Ensemble Framework

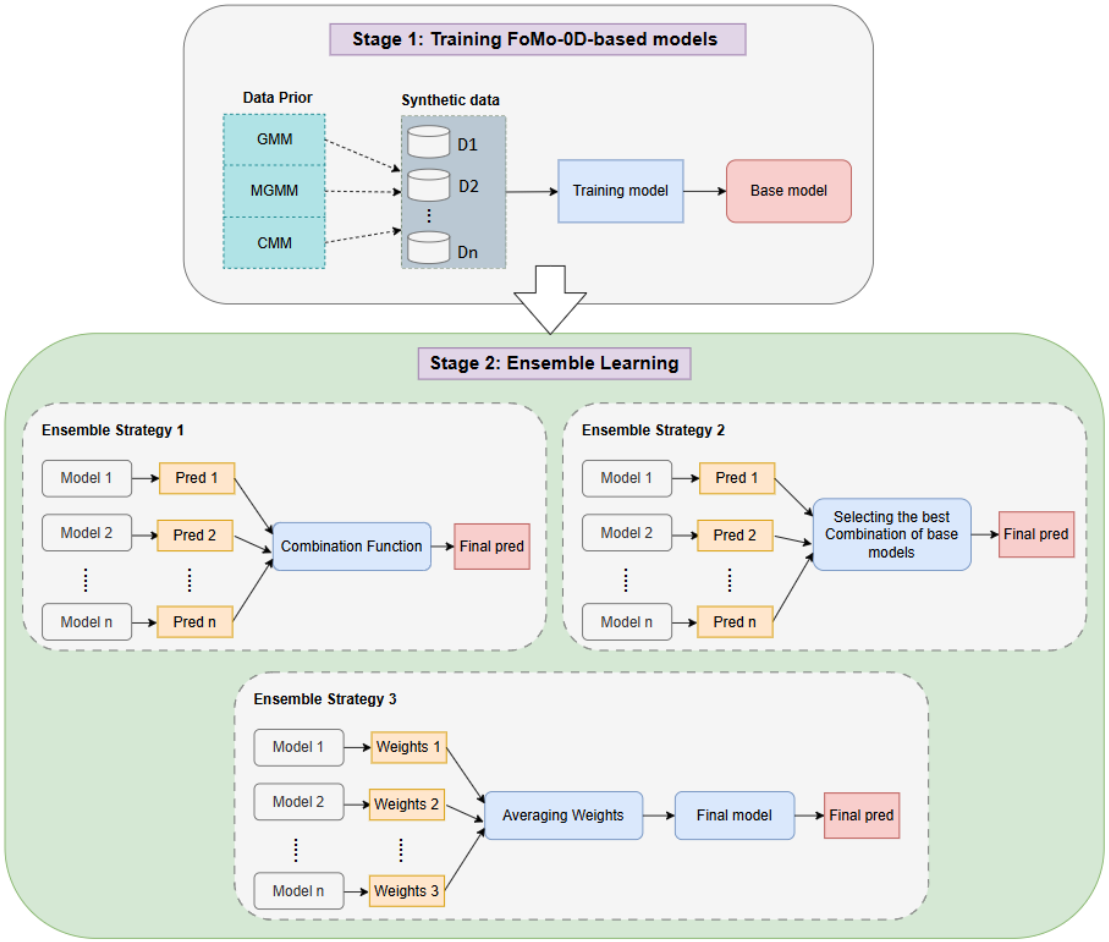


Figure 2: Overview of the proposed framework for constructing ensembles of foundation models

Ensemble Framework

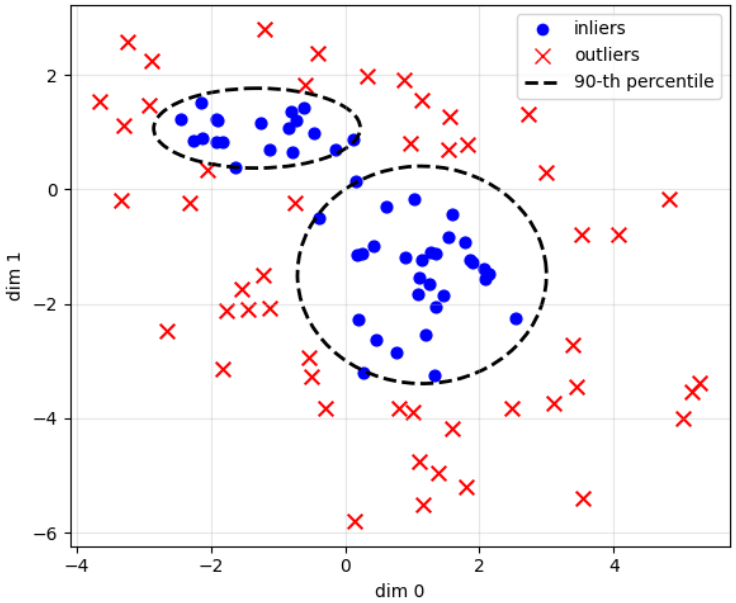
The first stage:

- Data Priors: Gaussian Mixture Models (GMMs), Modified Gaussian Mixture Models (MGMMs), and Cube Mixture Models (CMMs)
- Optimizing the FoMo-0D configuration

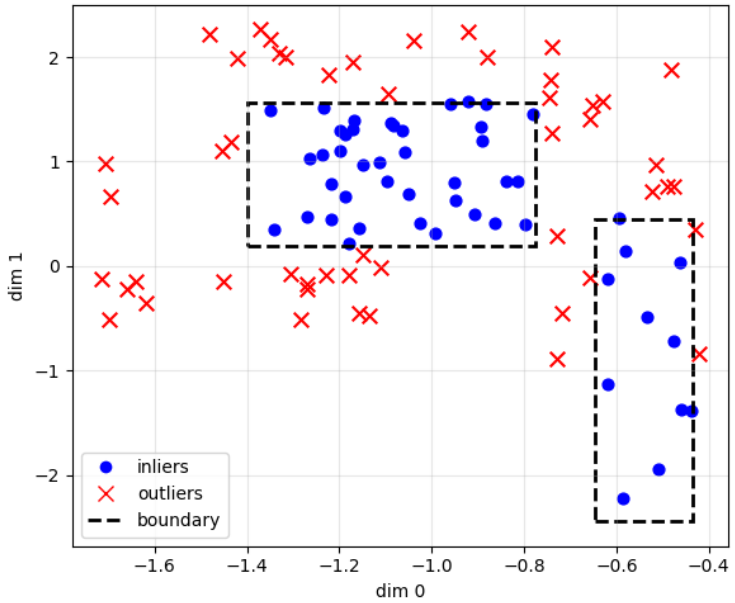
The second stage:

- Bagging: aggregating predictions from all base models
- Selected Bagging: evaluating different combinations of sub-models to select the best subset of base models before aggregation.
- Weight Average: averaging the weights of all sub-models to construct the final model.

Data Priors



(a) Synthetic data drawn from a MGMM.



(b) Synthetic data drawn from CMM.

Figure 3: Illustration of two-dimensional synthetic datasets drawn from a MGMM with the 90th percentile as the criterion and CMM.

Experiments and Evaluation

Experiment Setup

- **Benchmark Datasets**: 57 tabular outlier detection datasets from the ADBench benchmark [Han et al.(2022)].
- **Baseline Methods**: 27 baselines, including the FoMo-0D model and its 26 corresponding baselines.

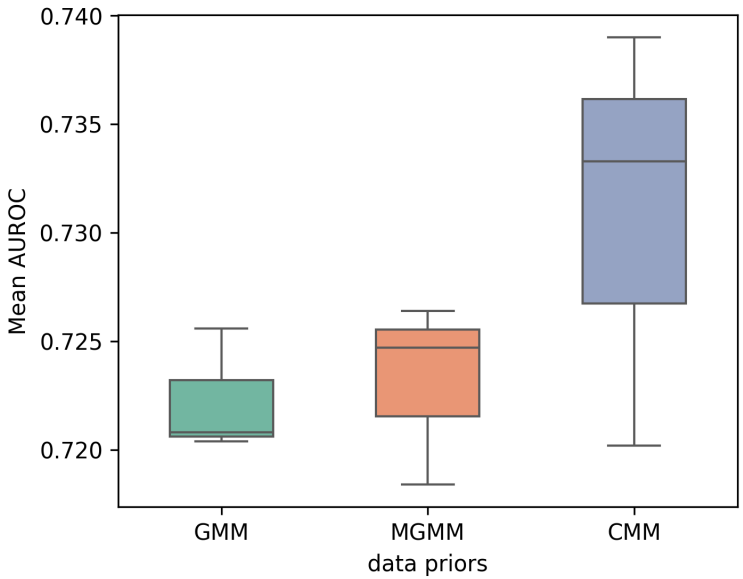
Experiment Setup

Hyperparameters for Optimization

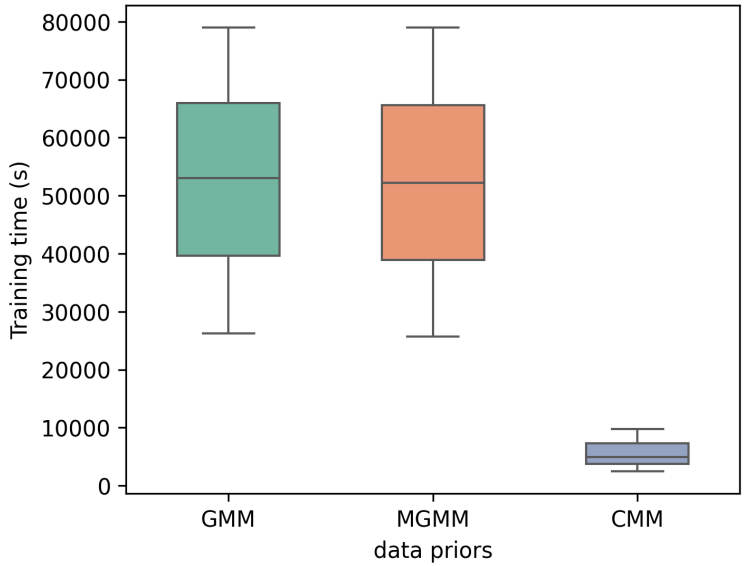
- Data priors: GMMs, MGMMs, and CMMs.
- max_feature_dim: The maximum number of features that a synthetic training dataset can have.
- max_num_cluster: The maximum number of clusters in a synthetic dataset.
- nlayer: The number of layers in the transformer encoder of the FoMo-0D architecture.
- nhead: The number of attention heads in the transformer encoder block.
- batch_size: The number of training datasets passed through a model in one iteration.
- epochs
- lr

1. Impact of Data Priors

9 base modes categorized into 3 groups according to their data prior.



(a) Mean AUROC scores of each model group



(b) Training time of each model group.

Figure 4: The influence of data priors on base models in terms of performance and training time.

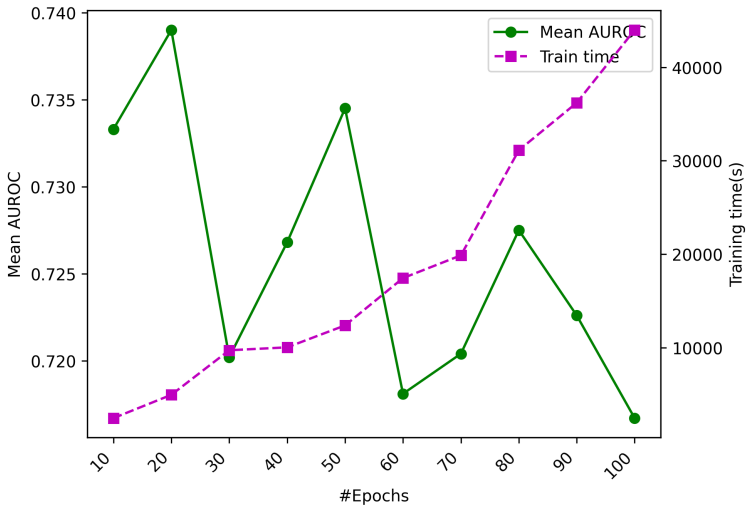
Results - Base Models

1. Impact of Data Priors

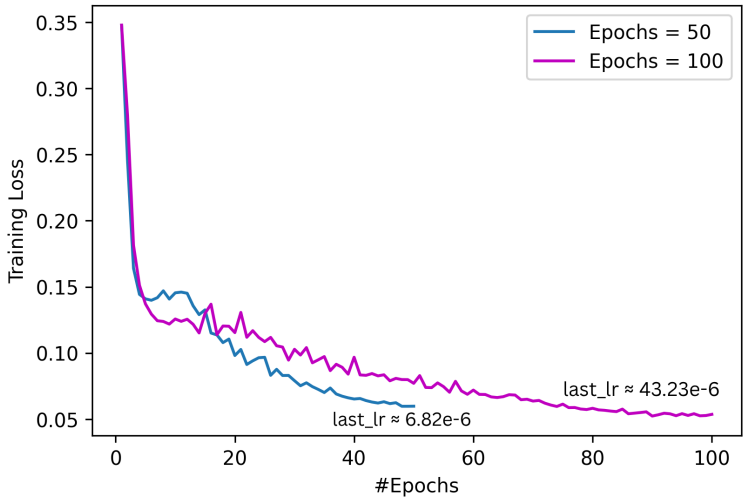
$\Rightarrow$ We use all three types of data priors to create a diversity of base models in an ensemble.

$\Rightarrow$ Sub-models trained on CMM-based synthetic datasets are used more than the others.

2. How many Epochs are enough?



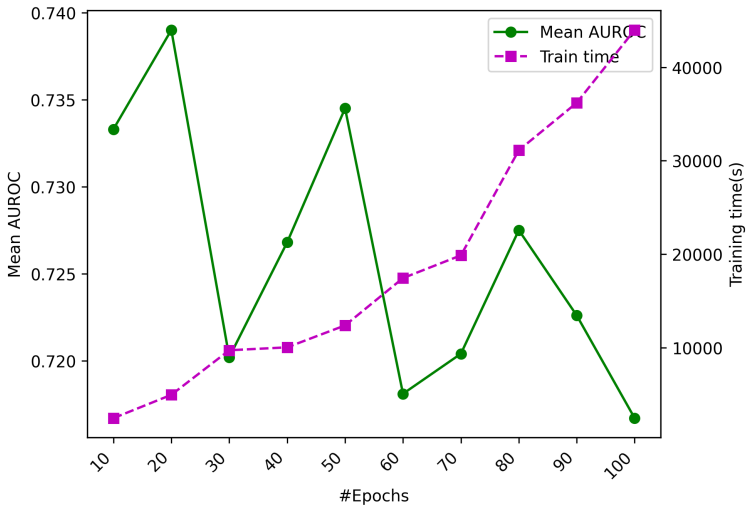
(a) Mean AUROC scores and Training time.



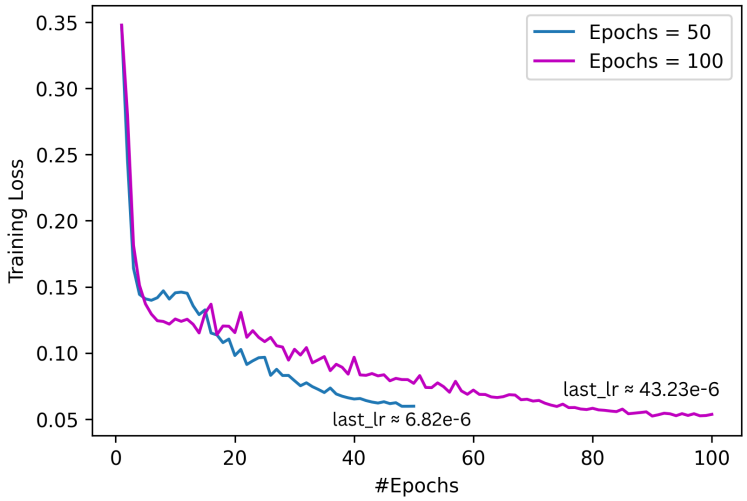
(b) Training losses and the final learning rates of two models.

Figure 5: The influence of the number of epochs on base models.

2. How many Epochs are enough?



(a) Mean AUROC scores and Training time.

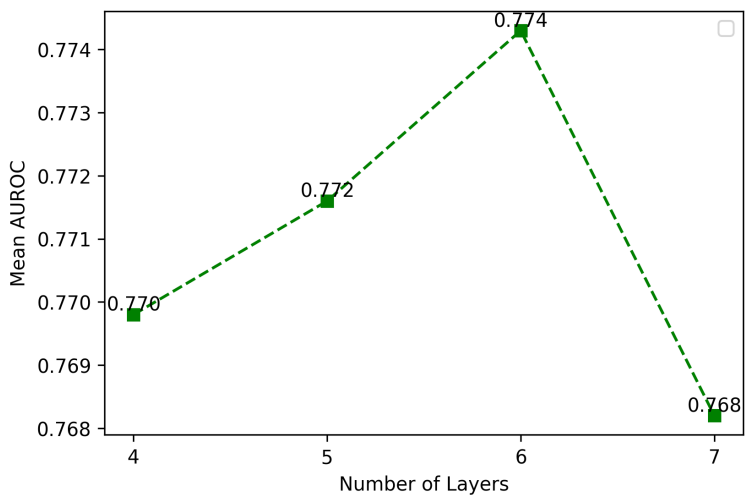


(b) Training losses and the final learning rates of two models.

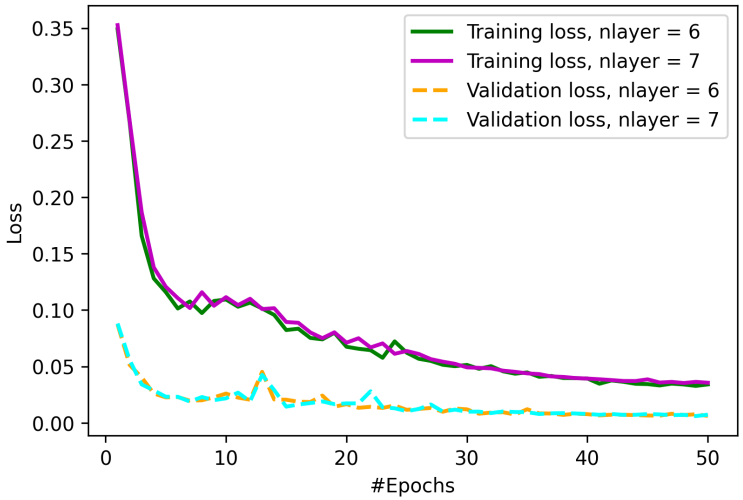
Figure 6: The influence of the number of epochs on base models.

⇒ We use epoch of 50 or even fewer for training base models.

3. Impact of nlayer



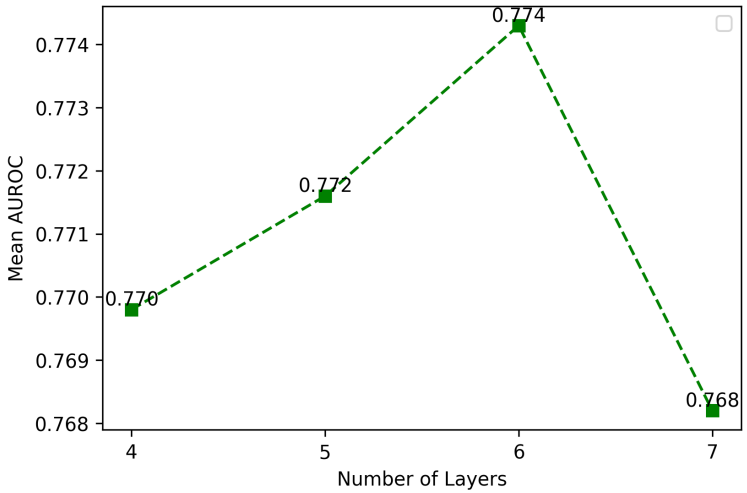
(a) Mean AUROC scores of models with various nlayer settings.



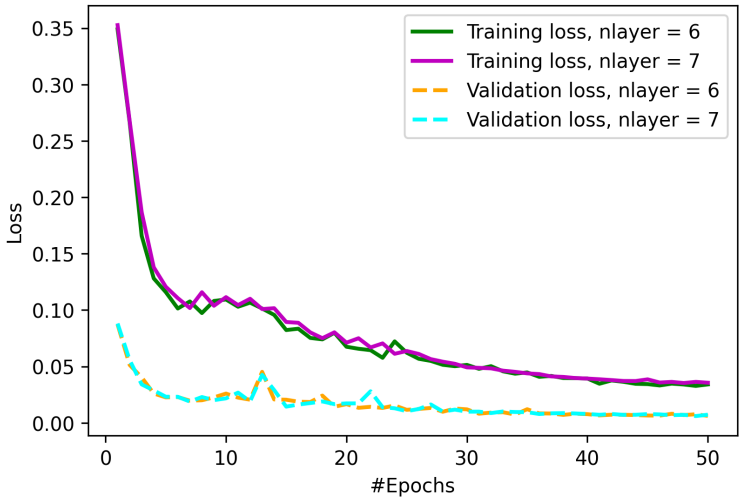
(b) Training and validation lossess of two models with nlayer of 6 and 7.

Figure 7: The impact of nlayer hyperparameter.

3. Impact of nlayer



(a) Mean AUROC scores of models with various nlayer settings.



(b) Training and validation lossess of two models with nlayer of 6 and 7.

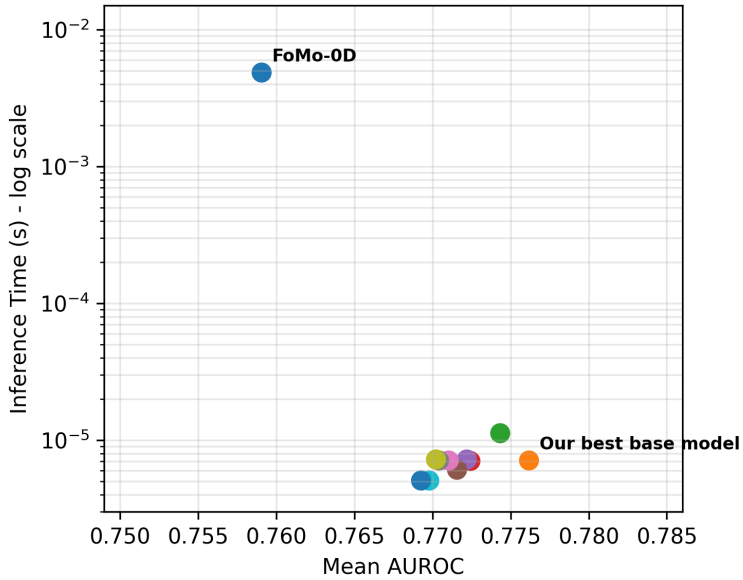
Figure 8: The impact of nlayer hyperparameter.

⇒ The reduction in performance is not due to classical overfitting
⇒ Increasing depth beyond a certain point may lead to poorer generalization

4. The FoMo-0D vs. best base models

Hyperparameter	FoMo-0D	Our best base model
max_feature_dim	100	70
max_num_cluster	5	70
batch_size	8	6
epochs	200	50
lr	0.001	0.001
nlayer	4	6
nhead	4	2

(a) The hyperparameter settings.



(b) Average inference time vs. mean AUROC scores.

Figure 9: Comparison between the original FoMo-0D and our top 10 best base models.

1. Which Ensemble Strategies outperform?

- Ensembles are formed by randomly sampling base models from the trained pool.
- The ensemble size is varied.

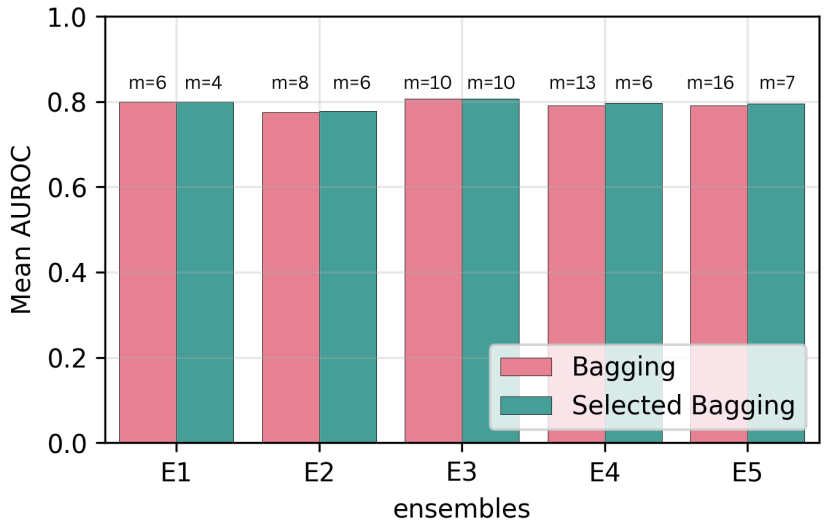


Figure 10: Comparison between Bagging and Selected Bagging strategies.

Results - Ensembles

1. Which Ensemble Strategies outperform?

- Both **Bagging** and **Selected Bagging** strategies maintain similar performance levels across different configurations.
- The Selected Bagging approach mostly achieves slightly better results while requiring fewer base models.
- Selected Bagging incurs significantly higher computational costs during training, but often requires less inference time.
- Due to infrastructural resource constraints, we do not extend our investigation of Selected Bagging to larger ensembles.

2. Do large ensembles perform better?

- Ensembles still benefit from an increasing number of base models.
- The selection of base models included in an ensemble also contributes to improvement of the ensemble generalization capacity.

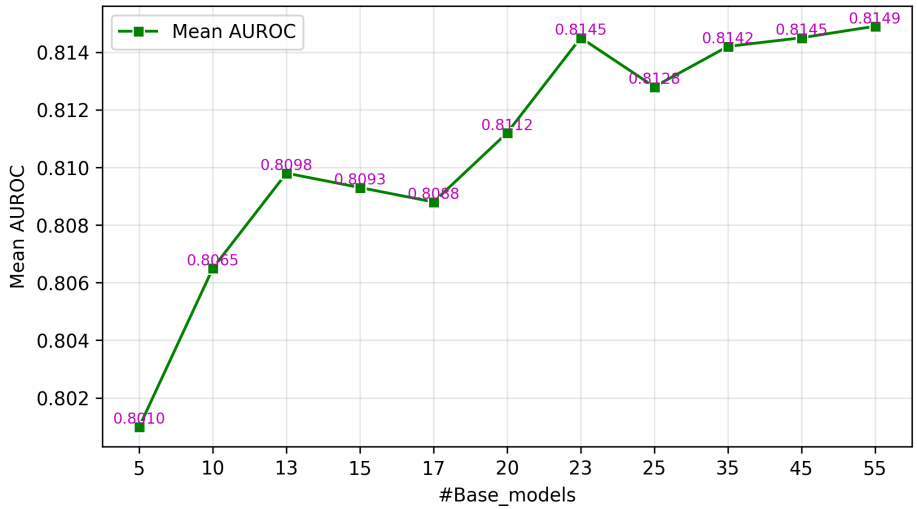


Figure 11: The impact of ensemble size on performance.

3. Impact of the diversity on ensemble performance

- The Diversity_5 and Diversity_10 ensembles outperform their respective homogeneous counterparts.

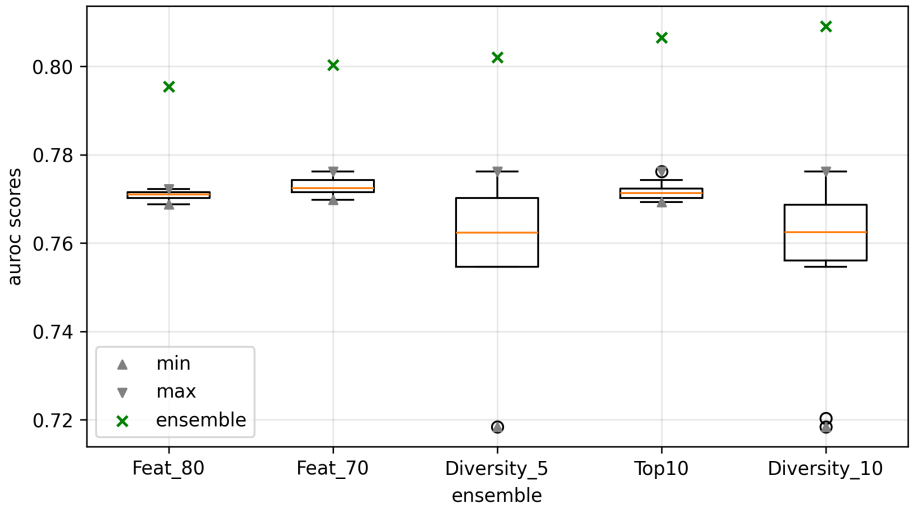


Figure 12: Boxplots show the distribution of AUROC scores of base models included in an ensemble, while green Markers show the AUROC scores of ensembles.

Results - Ensembles

3. Impact of the diversity on ensemble performance

$\Rightarrow$ Individual ensemble members do not need to be very accurate but should differ sufficiently from each other.

$\Rightarrow$ By combining the predictions of a diverse set of models, the overall error of the ensemble can be reduced, enhancing its generalization capacity.

4. Our top ensembles vs. 27 baseline methods

- The 1st-, 2nd-, and 3rd-Bagging ensembles contain 55, 45, and 23 base models, respectively.
- The 1st-Selected Bagging ensemble consists of 12 models selected from a candidate pool of 15.

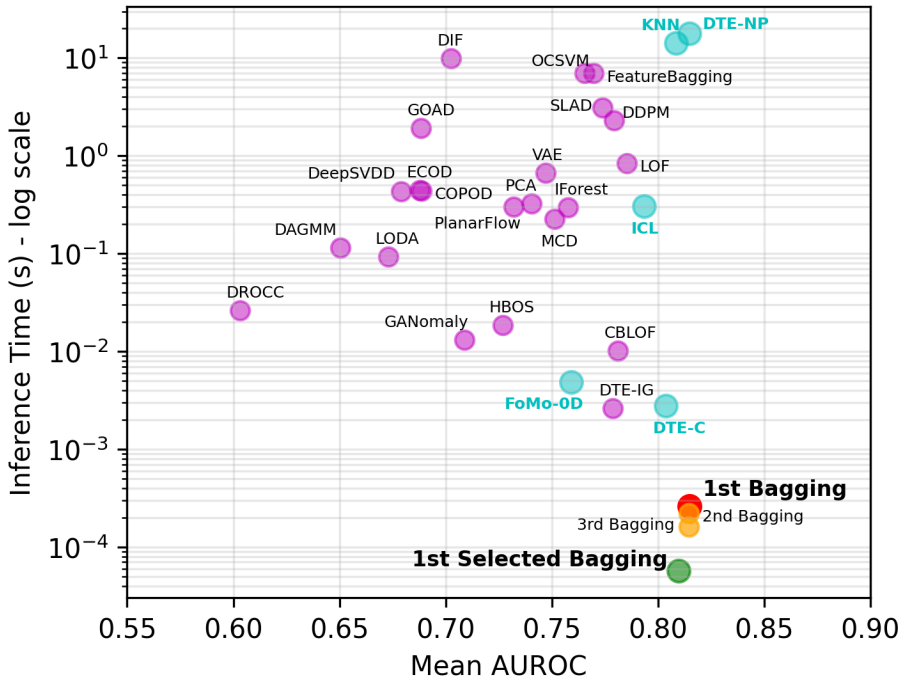


Figure 13: Average inference time vs. mean AUROC scores of our best ensembles compared to 27 baselines in the semi-supervised setting.

5. The two-sided Wilcoxon signed rank test

- Two-sided Wilcoxon signed rank tests between the **1st-Bagging ensemble**, and the FoMo-0D and the top-4 best-performing baselines.
- The null hypothesis: there is no significant difference between the paired performances.
- $\alpha = 0.05$

Test	p-value
DTE-NP	0.939
kNN	0.401
ICL	**0.043**
DTE-C	**0.012**
FoMo-0D	$\mathbf{1.281 \times 10^{-12}}$

Figure 14: p-values of the two-sided Wilcoxon signed rank test, comparing our **1st-Bagging ensemble** to the top-4 baseline methods and FoMo-0D.

Conclusion and future work

Conclusion

- Optimizing foundation models can reduce computational requirements while enhancing generalization capacity.
- Increasing the number of base models can enhance the ensemble performance, but diversity among the components is also crucial.
- Selecting suitable base models included in an ensemble can boost the generalization capability while reducing inference latency.
- Both **Bagging** and **Selected Bagging** strategies demonstrate substantial improvements in terms of performance and inference efficiency.

$\Rightarrow$ The results indicate the scalability, robustness, and strong generalization capability of foundation model ensembles in practice.

Future Work

- Training foundation models on a set of synthetic datasets generated from multiple data priors.
- Exploring weighted ensemble methods, where each base model is assigned a contribution weight based on its relative importance.
- Further investigating the **Selected Bagging** and **Weight Average** strategies at larger scale.

Thank you for listening

References



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Advances in neural information processing systems 35 (2022), 32142–32159.



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FoMo-0D: A Foundation Model for Zero-shot Outlier Detection. In *1st ICML Workshop on Foundation Models for Structured Data*.